

TRAVEL STYLE SEGMENTATION & DRIFT MONITORING

ITCS355 - Machine Learning Operations

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ONE PROFILE, A SIMPLE TRAVEL SEGMENT

A travel platform can use rating profiles to group similar interests.

INPUT

24 category ratings [1]

Each value: 0-5

MODEL

K-Means, k=5

Shared preprocessing

OUTPUT

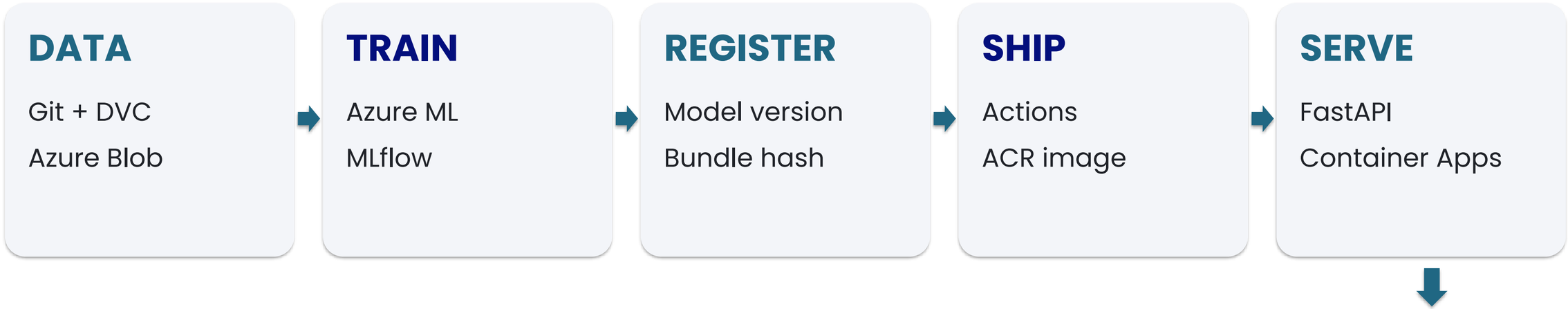
Segment + distance

Uncertainty + version

Segment names are provisional. This is a preference tool, not a personality test.

Source: docs/MODEL_CARD.md; reports/bundle-manifest.json; UCI [1]

FROM VERSIONED DATA TO A MONITORED API



Tests gate deployment.

Failure checks feed back into tests.

Azure region: centralindia | Provider code: cloudlayer/

Source: README.md; docs/reference/course/project/project-brief.md [2]

ONE VERSIONED DATASET, ONE REPEATABLE PIPELINE

5,456
Raw rows

5,454
Accepted rows

2
Quarantined rows



CC BY 4.0 [1] | Raw SHA-256 + DVC pointer | Seed 42

Train-only imputation/scaling. Identical profiles stay in one split.

Source: reports/data-audit.json; prepared-data.json; selection.json; UCI [1]

THE SAME BUNDLE REACHES THE API

MANAGED TRAINING

Azure ML job: Completed

Fresh-fit equivalence: passed

Registry: travel-style, version 1

SERVING BUNDLE

Preprocessing + feature order

Centers + mapping + thresholds

Hash and golden checks at startup

Model version: abfd88e893ca

Immutable images use one canonical bundle.

```
Bundle SHA-256: 71f3ba41b4b964f5...
CI image digest: f124df59007a...
```

Source: reports/finish-training-registry.json; bundle-manifest.json; ci-deploy-result.json

LIVE DEMO: VALID AND UNEXPECTED INPUT

Request	Expected / verified
Valid 24-field profile	200 + segment
Missing fields / invalid rating	422, clear validation error
Batch of 100 profiles	200; golden check passed
Semantic canary	200; predictions match

SAVED REAL HTTP RESPONSE


```
{
  "segment": "Segment 1",
  "cluster_id": 0,
  "distance": 4.1435,
  "uncertain": false,
  "model_version": "abfd88e893ca"
}
```


Exactly 24 named, finite ratings in [0,5]. A valid uncertain profile returns a null segment.

Source: output/presentation/evidence/api-demo.json; service/schemas.py

FAILED TESTS BLOCK DEPLOYMENT

NORMAL RUN

Tests: success

Deploy: success

Real HTTP smoke: passed

Open GitHub Actions run 38173375457

DELIBERATE FAILURE

Tests: failure

Deploy: skipped

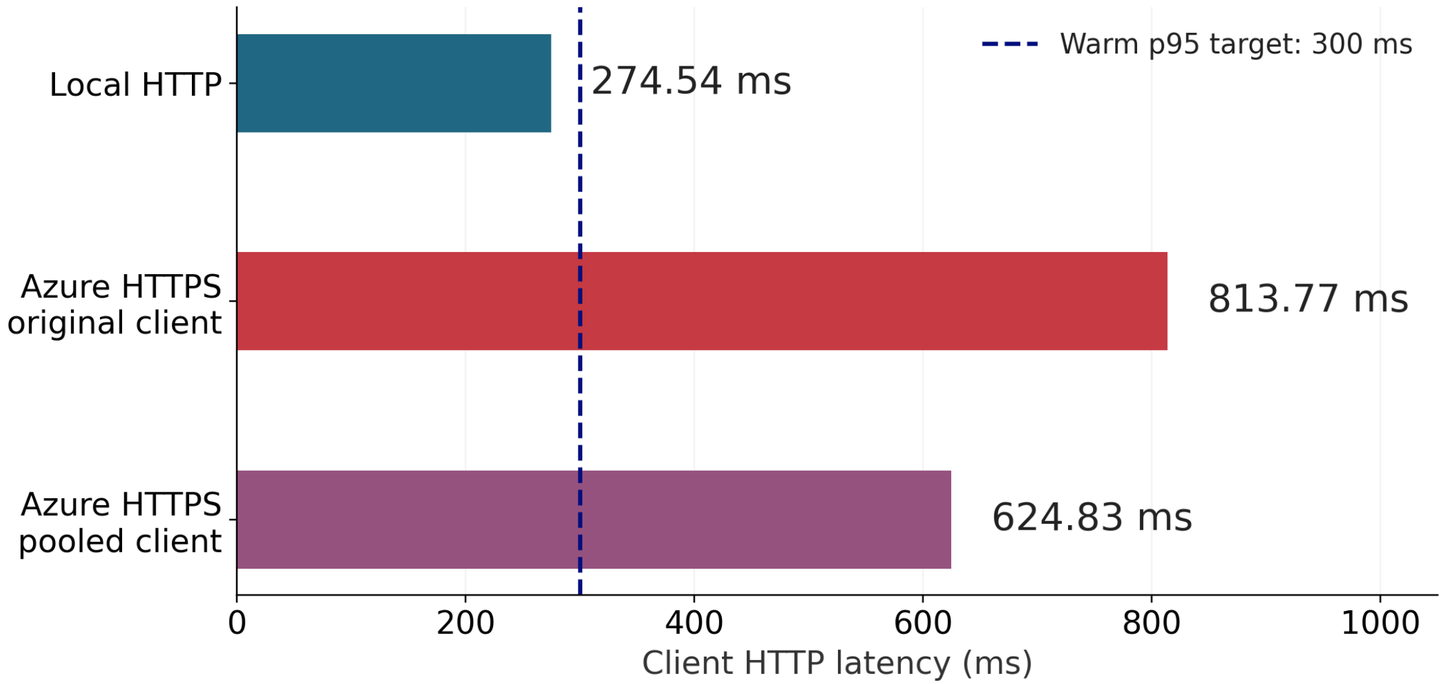
Failed code was not shipped

Open GitHub Actions run 38173380800

Same tested source commit: 9fea51a | OIDC login + limited roles | Immutable image

Source: reports/ci-runs.json; reports/ci-deploy-result.json; .github/workflows/ci.yml

THE API WORKS; CLOUD LATENCY MISSES OUR TARGET



WARM TEST

200 requests

Concurrency: 10

Errors: 0 in all runs

COLD REQUEST

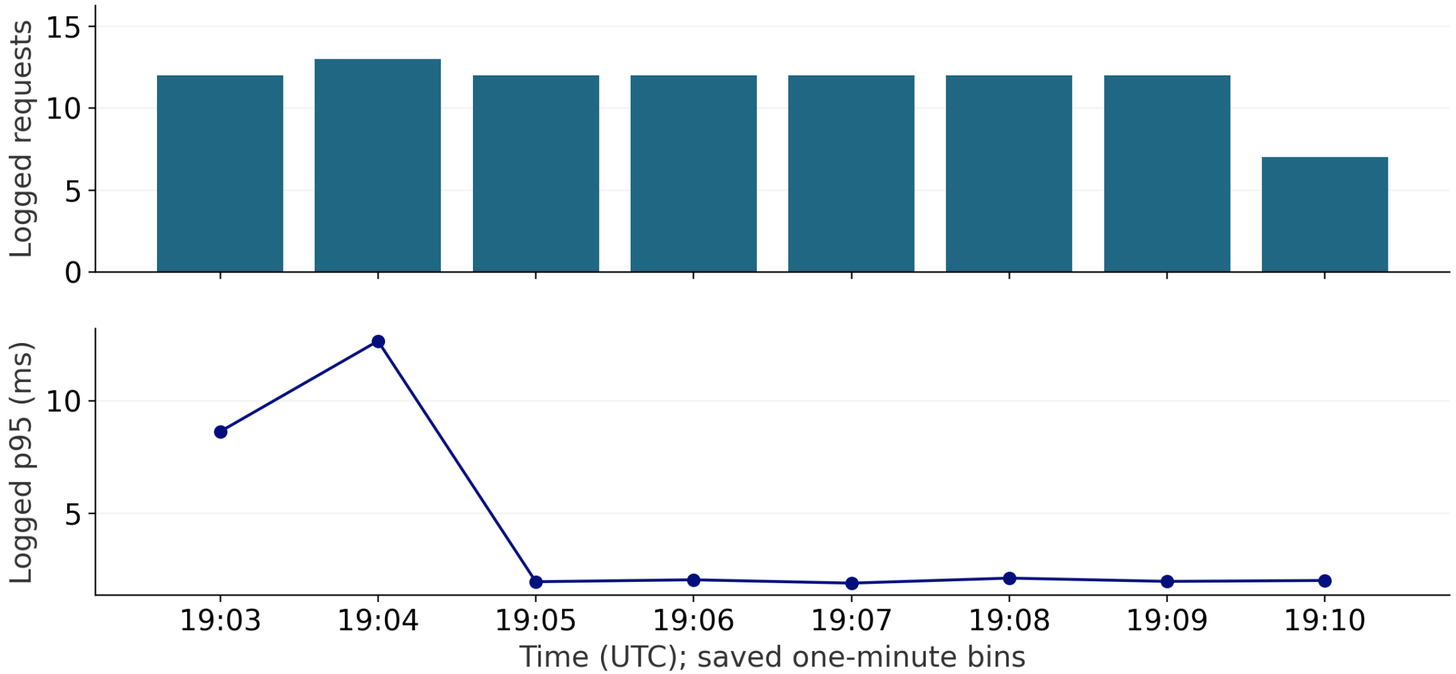
3.82 seconds

After verified zero replicas

One WSL HTTPS sample

Source: reports/serving-recheck/local-load.json; azure-load.json; pooled-load.json; finish-cold.json

MONITOR REQUESTS AND CHECK PREDICTION MEANING



DASHBOARD

MLOP Travel Style

Requests, errors, latency

Model version in logs

ACTUAL ALERT

Semantic failure detected

Alert fired, then resolved

Email is not configured

Saved Log Analytics results; application timing, not client HTTPS latency.

Source: reports/azure-telemetry.json; finish-monitor.json; finish-alert.json

A WRONG SEGMENT MAPPING TRIGGERED DETECTION AND RECOVERY

Controlled fault in the separate staging app: wrong cluster-to-segment mapping.

INJECT

16:54:22 UTC

Wrong mapping

DETECT

16:54:52 UTC

Canary: 503

ALERT

16:56:39 UTC

Actual alert fired

RECOVER

17:02:08 UTC

Canary: 200

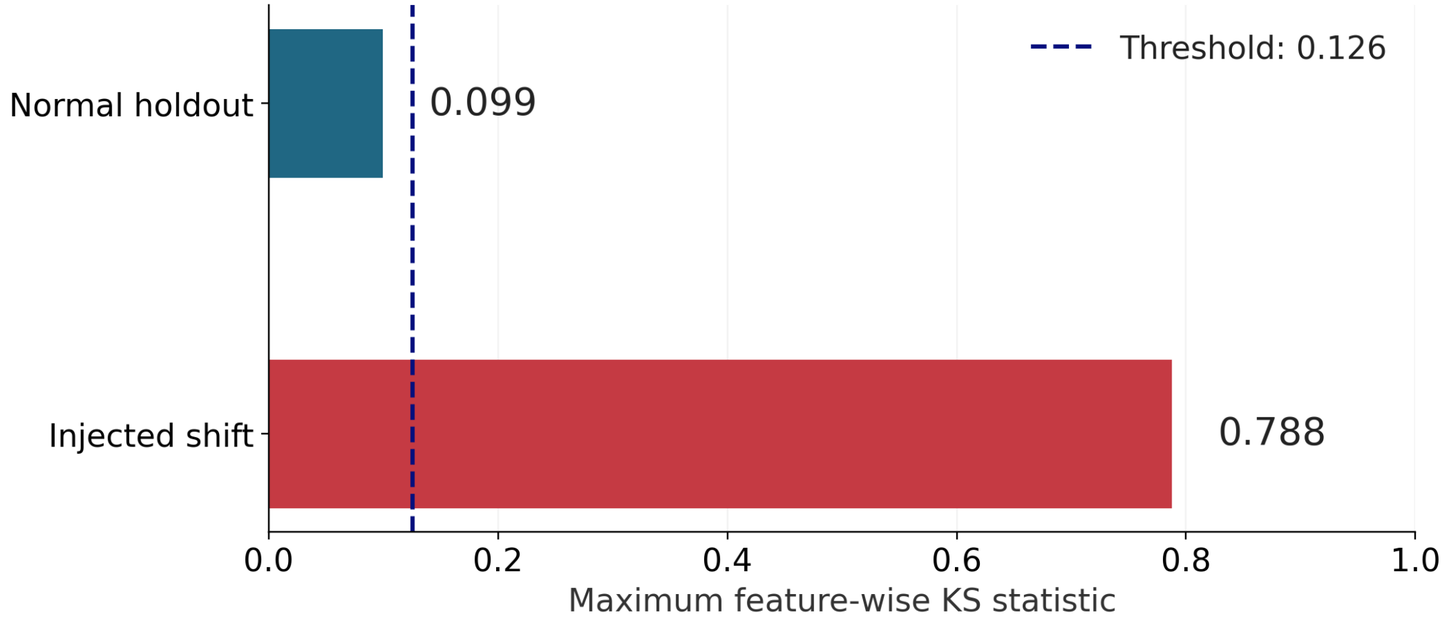
Detection to confirmed recovery: 7 min 16.4 sec, including the failed first recovery attempt.

Fix: restore config, restart staging, confirm the canary twice, and keep a regression test.

Saved failure evidence. Arrows show event order, not equal time gaps.

Source: reports/finish-failure.json; finish-alert.json; tests/test_api.py

A CONTROLLED SHIFT IS DETECTED OFFLINE



CONTROLLED TEST

Reference: 1,908 rows

Recent sample: 200 rows

100 calibration rounds

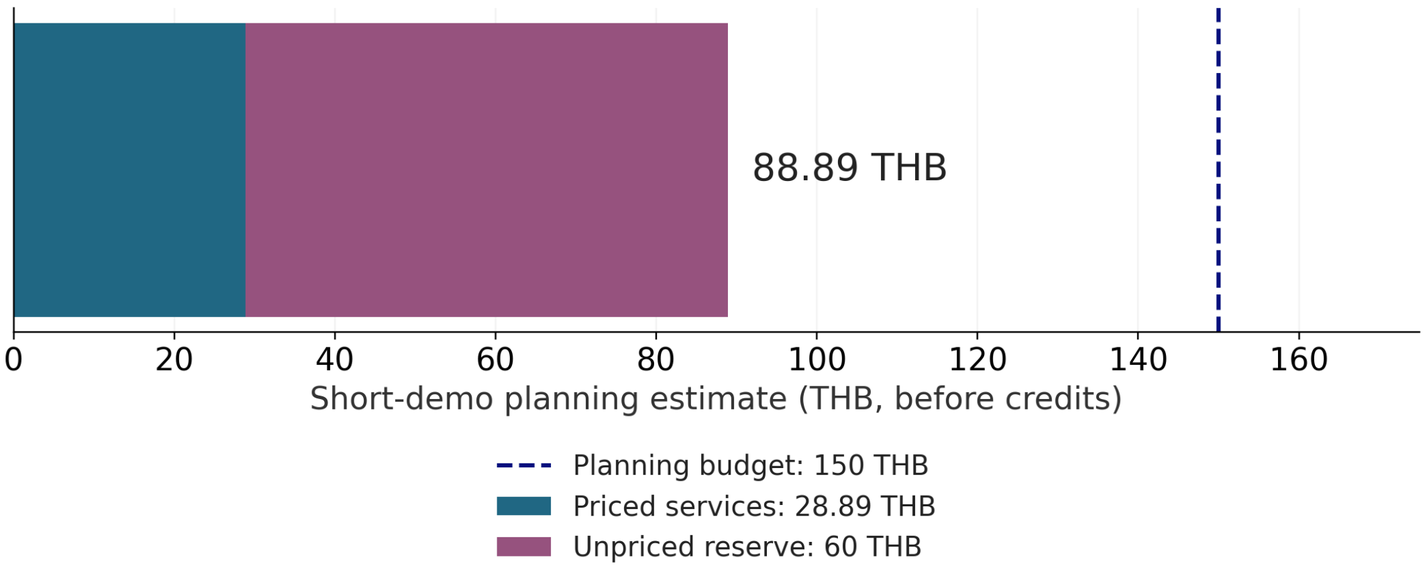
Shift: +1.5 in four ratings, clipped to [0,5].

Offline input drift only.

No concept-drift claim.

Source: reports/input-drift.json; monitoring/drift.py

COST PER 1,000 PREDICTIONS: THE ESTIMATE AND ITS LIMITS



PLANNED COST

8.89 THB / 1,000

Assume 10,000 predictions

One prediction per request

REPORTED COST

0.00143104 USD / 1,000

841 observed profiles

Provisional; not a final bill

Assumptions: 40 THB/USD; training 1 h; app active 2 h; ACR 2 days. No credits applied.

Source: reports/azure-cost-plan.json; cost-per-1000.json; docs/COST_REPORT.md

WHAT ANOTHER WEEK WOULD IMPROVE

Priority	Next action
Cloud warm latency	Measure network vs. application time; retest p95
Cost reconciliation	Recheck full cost and usage when billing updates
Live feature drift	Add rolling windows when current data is available
User quality	Review segment labels and test useful recommendations

Current limits: provisional labels, unknown zero meaning, and no user-quality study.

Save evidence before cleanup. Remove cloud resources after the final demo.

Source: reports/finish-delivery.json; docs/DELIVERY_STATUS.md; docs/FINAL_STEPS.md

REFERENCES

[1] S. Renjith, "Travel Review Ratings," UCI Machine Learning Repository, 2018.
DOI: 10.24432/C5C31Q. License: CC BY 4.0.

[2] ITCS355, "Capstone Project Brief," course/project/project-brief.md.
Presentation: 8 minutes + 5 minutes of questions.

[3] MLOP-PJ team, "Project reports and CI runs."
Data, registry, API, load, alert, recovery, drift, and cost evidence.

Dataset: https://doi.org/10.24432/C5C31Q

Repository: https://github.com/DewPerawis/MLOP-PJ (access required)

Source: UCI dataset page; course brief; project reports and CI runs

THANK YOU